

Cancer chemopreventive potential of apples, apple juice, and apple components.



Apples (*MALUS* sp., Rosaceae) are a rich source of nutrient as well as non-nutrient components and contain high levels of polyphenols and other phytochemicals. Main structural classes of apple constituents include hydroxycinnamic acids, dihydrochalcones, flavonols (quercetin glycosides), catechins and oligomeric procyanidins, as well as triterpenoids in apple peel and anthocyanins in red apples. Several lines of evidence suggest that apples and apple products possess a wide range of biological activities which may contribute to health beneficial effects against cardiovascular disease, asthma and pulmonary dysfunction, diabetes, obesity, and cancer (reviewed by Boyer and Liu, Nutr J 2004). The present review will summarize the current knowledge on potential cancer preventive effects of apples, apple juice and apple extracts (jointly designated as apple products). In brief, apple extracts and components, especially oligomeric procyanidins, have been shown to influence multiple mechanisms relevant for cancer prevention in IN VITRO studies. These include antimutagenic activity, modulation of carcinogen metabolism, antioxidant activity, anti-inflammatory mechanisms, modulation of signal transduction pathways, antiproliferative and apoptosis-inducing activity, as well as novel mechanisms on epigenetic events and innate immunity. Apple products have been shown to prevent skin, mammary and colon carcinogenesis in animal models. Epidemiological observations indicate that regular consumption of one or more apples a day may reduce the risk for lung and colon cancer.